

Bhathiya Hemanthage

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Education

PhD in Robotics and Autonomous Systems

Sep 2020 – Mar 2026

Thesis: Generalized Visual-Language Grounding with Complex Language Context
Heriot-Watt University & University of Edinburgh (Edinburgh Centre for Robotics)
Supervisor: Prof. Oliver Lemon

M.Sc.(Major Component of Research)

May 2019 – Aug 2020

Thesis: Dialogue State Tracking for Low Resource Languages
University of Moratuwa
Supervisor: Dr. Uthayasanker Thayasivam

B.Sc. (Eng) Hons. Computer Science Engineering

Sep 2011 – Apr 2016

University of Moratuwa
Second Class Honours Upper Division

Publications

Conference Papers

1. **RECANTFormer: Referring Expression Comprehension with Varying Numbers of Targets**
Hemanthage B., Bilen H., Bartie P., Dondrup C., Lemon O.
2024 Annual Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024
2. **Divide and Conquer: Rethinking Ambiguous Candidate Identification in Multimodal Dialogues with Pseudo-Labelling**
Hemanthage B., Dondrup C., Bilen H., Lemon O.
25th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDial), 2024
3. **Multitask Multimodal Prompted Training for Interactive Embodied Task Completion**
Pantazopoulos G., Nikandrou M., Parekh A., **Hemanthage B.**, Eshghi A., Konstas I., Rieser V., Lemon O., Suglia A.
2023 Annual Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023
4. **A Simple Language Model for Multimodal Task-Oriented Dialogue with Symbolic Scene Representation**
Hemanthage B., Bartie P., Dondrup C., Lemon O.
15th International Conference on Computational Semantics (IWCS), 2023
5. **ViCA: Combining Visual, Social, and Task-Oriented Conversational AI in a Healthcare Setting**
Pantazopoulos G., Bruyere J., Nikandrou M., Boissier T., **Hemanthage B.**, Binh S., Shah V., Dondrup C., Lemon O.
23rd ACM International Conference on Multimodal Interaction (ICMI), 2021
6. **Meta Learning for Few-shot Joint Intent Detection and Slot Filling**
Bhathiya H., Thayasivam U.
5th International Conference on Machine Learning Technologies (ICMLT), 2020
7. **Workload and Resource Aware Proactive Auto-scaler for PaaS Cloud**
Shariffdeen, R., Munasinghe D., **Bhathiya H.**, Bandara U. Bandara H.
IEEE 9th International Conference on Cloud Computing (CLOUD), 2016
8. **Adaptive workload prediction for proactive auto scaling in PaaS systems**
Shariffdeen, R., Munasinghe D., **Bhathiya H.**, Bandara U. Bandara H.
2nd International Conference on Cloud Computing Technologies and Applications (CloudTech)

Papers in Progress

1. **CRAFTeD: Benchmark for Compositional Reasoning in Free-form Text Object Detection**
Hemanthage B., Lemon O., Bilen H.

Demo and Workshop Papers

1. **Generalized Visual-Language Grounding with Complex Language Context**
Hemanthage B.
20th Workshop of Young Researchers' Round table on Spoken Dialogue Systems (YRRSDS), 2024
2. **Demonstrating EMMA: Embodied MultiModal Agent for Language-guided Action Execution in 3D Simulated Environments**
Suglia A., *Hemanthage B.*, Pantazopoulos G., Nikandrou M., Parekh A., Eshghi A., Konstas I., Rieser V., Lemon O.
23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL), 2022
3. **GLIMMeR: Global-Local Information-aware Multimodal grounding with GPT for Co-reference Resolution**
Hemanthage B., Bartie P., Dondrup C., Lemon O.
AAAI 2022 DSTC10 Workshop, 2021

Project Reports

1. **EMMA: A foundation model for embodied, interactive, multimodal task completion in 3D environments**
Parekh A.*, Pantazopoulos G.*, Nikandrou M.*, *Hemanthage B.**, Eshghi A., Konstas I., Lemon O., Suglia A., Alexa Prize SimBot Challenge Proceedings, 2023

Research Projects

Alexa Prize SimBot Challenge	<i>Jan 2022 – Apr 2023</i>
Member of Team EMMA, representing Heriot-Watt University (Finalist)	
◦ Developed an interactive agent capable of following user instructions for task completion while engaging in dialogue to clarify ambiguous directives.	
Situated Interactive Multimodal Conversation (SIMMC) 2.0 Challenge	<i>Jun 2021 – Dec 2022</i>
◦ Developed a GPT-2 based model focused on integrating global and local information within multimodal conversational agents.	
◦ Placed first in the Multimodal Response Generation sub-task and secured runner-Up in the Multimodal Disambiguation sub-task.	

Industry Experience

Senior Software Engineer	<i>Colombo, Sri Lanka</i>
<i>CodeGen International</i>	<i>Apr 2016 – May 2019</i>
◦ Designed and developed advanced chatbots specialized in negotiating hotel, flight, and cruise bookings, tailored to meet the specific needs of travel agency clients	
Software Engineering Intern(Cloud Team)	<i>Colombo, Sri Lanka</i>
<i>WSO2</i>	<i>Oct 2014 – Apr 2015</i>
◦ Designed and developed Apache jclouds support for Azure cloud service,	
Google Summer of Code Student	<i>Apr 2014 – Aug 2014</i>
◦ Implemented Google Cloud Storage support for the Apache jclouds open source project	

Teaching Experience

Statistical Machine Learning	<i>Heriot-Watt University</i>
<i>Enhanced Lab Helper</i>	<i>Sep 2022 – Jan 2023</i>

Databases and Information Systems*Enhanced Lab Helper**Heriot-Watt University**2022 – 2023***Machine Learning***Teaching Assistant**University of Moratuwa**Nov 2019 – May 2020***Data Structures and Algorithms***Teaching Assistant**University of Moratuwa**Jun 2019 – Oct 2019***Data Science Concepts***Visiting Instructor**CICRA Campus**Oct 2019 – Feb 2020***Technical Skills**

Programming

Python, Java

ML Frameworks

PyTorch, PyTorch Lightning

DevOps

Docker, Kubernetes, Slurm, Git